



IBM Data Analyst Capstone Project:

Analysis on Emerging
Technology Skills and Trends

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EXECUTIVE SUMMARY



- This report analyzes current and future technology trends through interactive dashboards covering programming languages, databases, platforms, web frameworks, and demographics.
- The analysis shows that JavaScript is the most widely used programming language and remains the most desired language for future work. HTML/CSS and Python also rank among the top currently used languages.
- For databases, PostgreSQL is the most used database and shows a slight increase in projected future demand.
- Demographic insights indicate that the majority of respondents are from the USA, primarily within the 25–34 age group, and most hold a Bachelor's degree.
- Overall, the results demonstrate strong alignment between current technology usage and future demand, highlighting the continued relevance of JavaScript and PostgreSQL in the IT industry.





INTRODUCTION



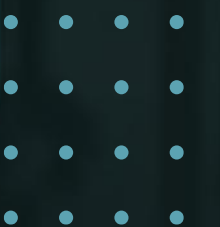
This report uses data analytics to highlight current and future trends in skills related to programming languages, databases, platforms, and web frameworks.

Key questions investigated:

- Which programming languages are most in demand today?
- What are the most in-demand database skills?
- Which IDEs or web frameworks are most popular?

Target Audience:

- IT professionals, HR managers, and anyone interested in the IT sector who wants to learn about top in-demand skills that will remain relevant in the future





METHODOLOGY



Data Source & Collection

- Data obtained from an online survey dataset [[link](#)].
- Collected additional information from websites using web scraping and API requests with Python's Requests library.



Exploratory Data Analysis

- Analyzed data distributions and identified outliers.
- Examined correlations between key variables.



Data Wrangling

- Cleaned the dataset by removing unnecessary separators and handling missing values (either removed or replaced based on defined criteria).
- Ensured data consistency and readiness for analysis.



Data Visualization & Dashboards

- Created charts and graphs to highlight distributions, relationships, and comparisons.
- Developed interactive dashboards to present insights in a clear and user-friendly manner.

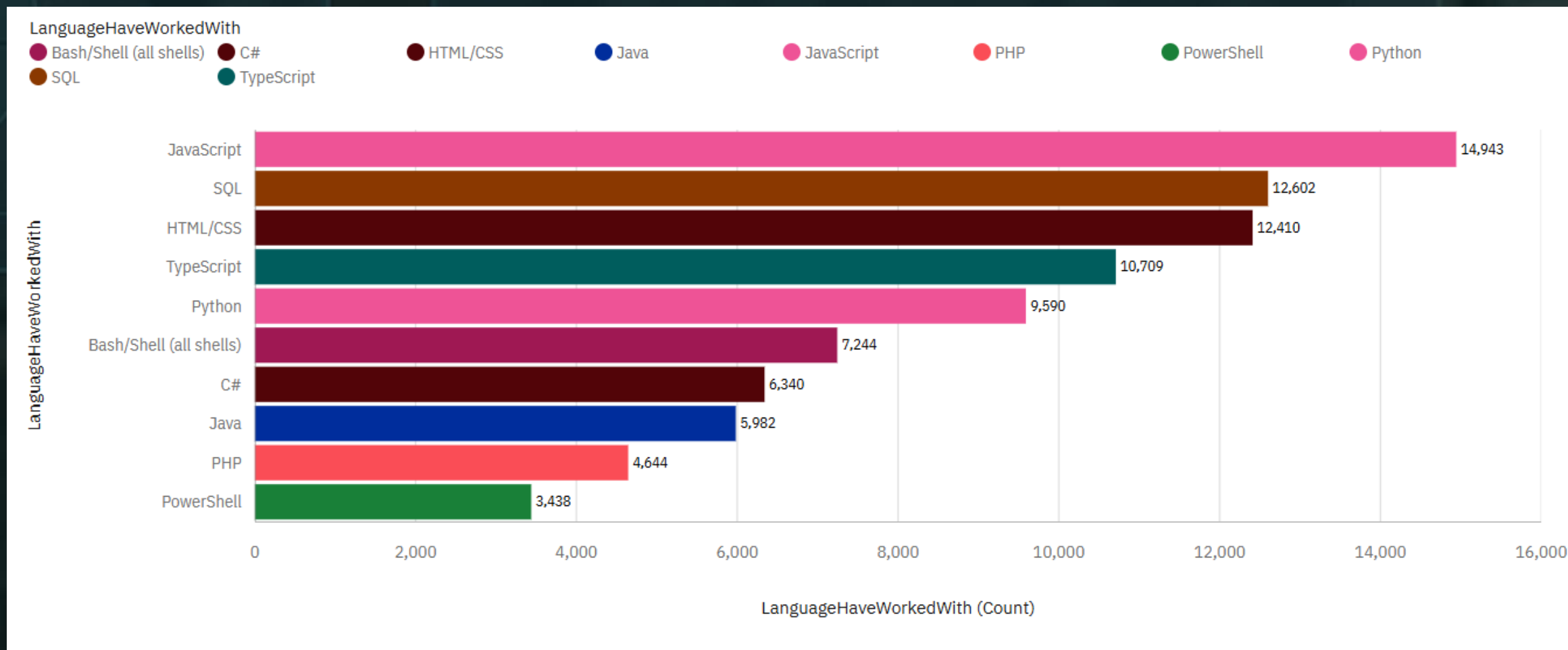


Results



PROGRAMMING LANGUAGE TRENDS

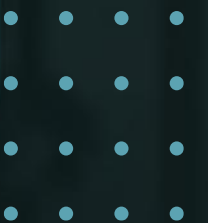
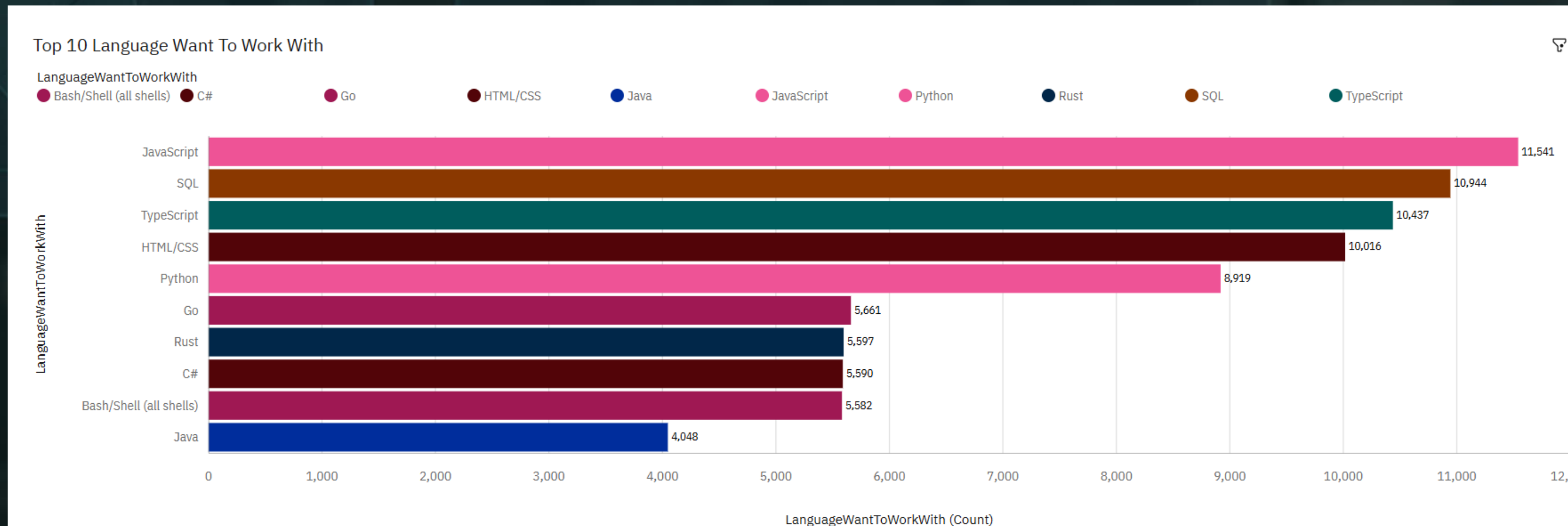
- Current Year





PROGRAMMING LANGUAGE TRENDS

- Next Year





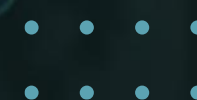
PROGRAMMING LANGUAGE TRENDS – FINDINGS & IMPLICATIONS

Findings:

- JavaScript and SQL maintain the top two positions in both the current and next year, showing strong and consistent demand.
- Web technologies (JavaScript, TypeScript, HTML/CSS) dominate the rankings, with TypeScript moving up next year.
- Python remains stable within the top 5 in both years.
- Emerging languages like Go and Rust enter the top 10 next year, replacing PHP and PowerShell.

Implications:

- The continued dominance of web technologies indicates sustained demand for web development skills.
- The rise of TypeScript reflects a shift toward more scalable and structured development practices.
- The entry of Go and Rust suggests growing interest in modern, high-performance programming languages.
- The stable position of SQL and Python highlights their ongoing importance across data and backend-related roles.

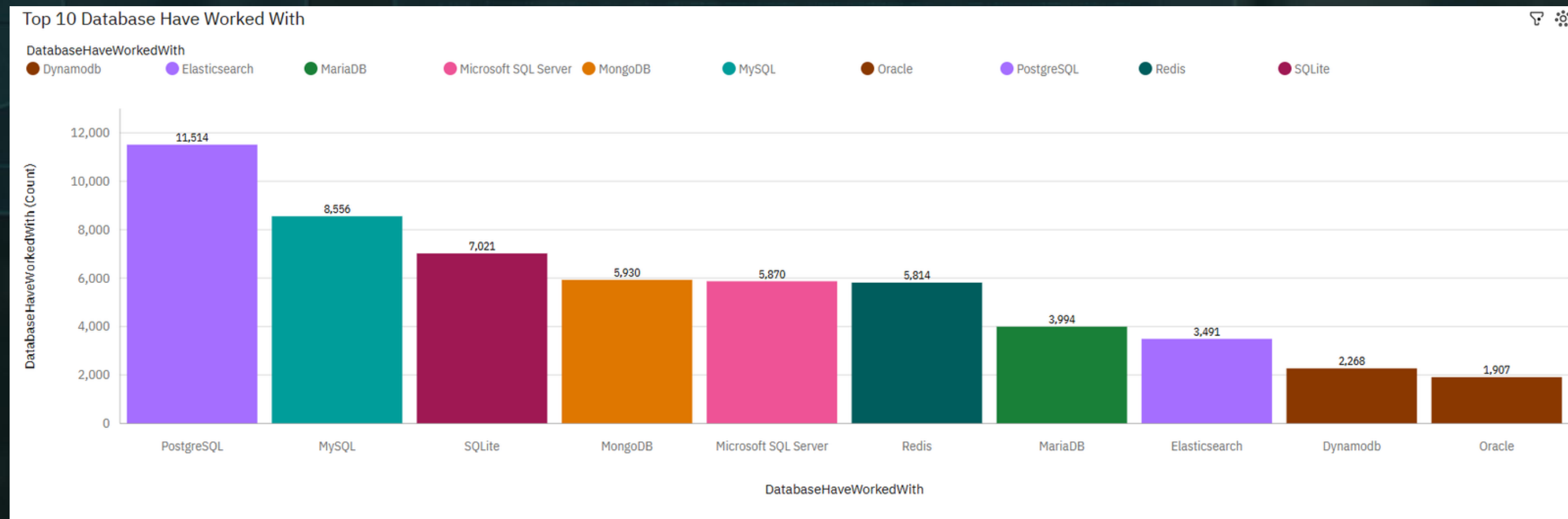




DATABASE TRENDS

- **Current Year**

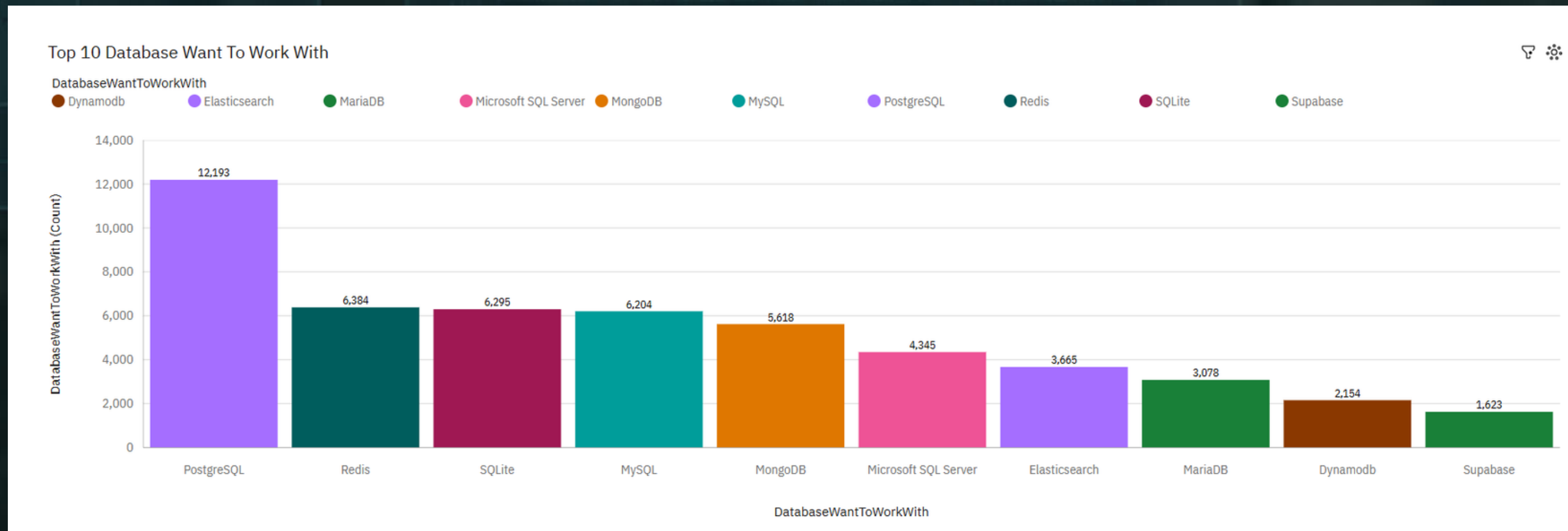
- **Next Year**





DATABASE TRENDS

- **Next Year**





DATABASE TRENDS – FINDINGS & IMPLICATIONS

Findings:

- PostgreSQL ranks first in both the current and next year, showing strong and consistent demand.
- Redis rises significantly from 6th place to 2nd place next year.
- MySQL drops from 2nd to 4th place, while SQLite maintains a strong top-three position.
- Supabase enters the top 10 next year, replacing Oracle.

Implications:

- The continued dominance of PostgreSQL indicates growing preference for advanced open-source relational databases.
- The strong rise of Redis suggests increasing demand for high-performance and caching solutions.
- The slight decline of MySQL may reflect a shift toward more feature-rich or scalable database systems.
- The entry of Supabase highlights growing interest in modern backend-as-a-service and cloud-based database solutions.



DASHBOARD

- The permanent link of the Cognos dashboard:

https://eu2.ca.analytics.ibm.com/bi/?perspective=dashboard&pathRef=.my_folders%2FNew%2Bdashboard&action=view&mode=dashboard&subView=model0000019ca19a40f7_00000007&nav_filter=true

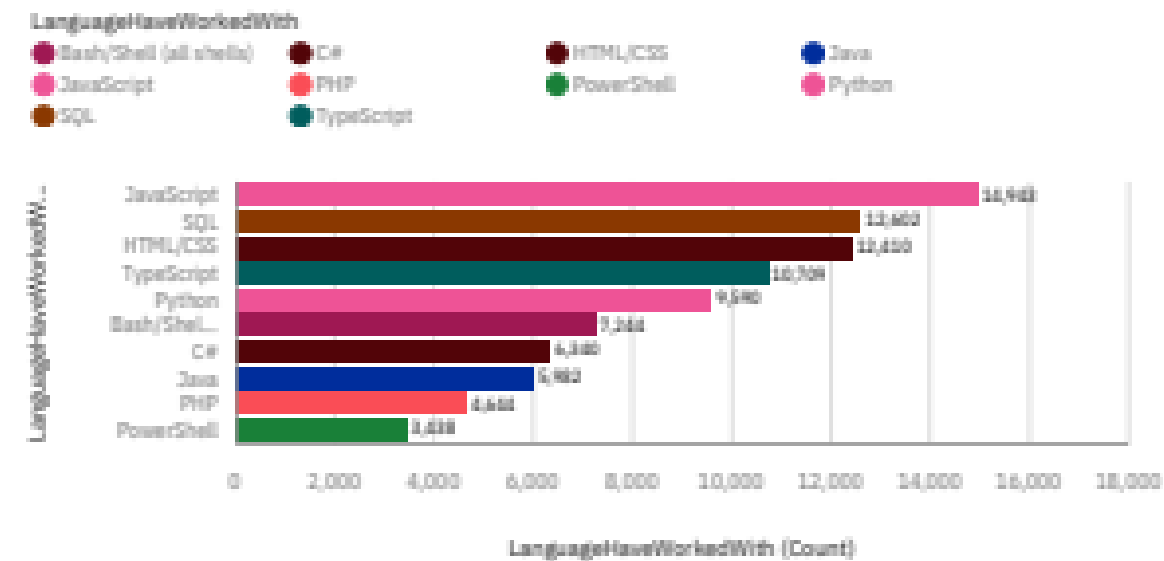


DASHBOARD

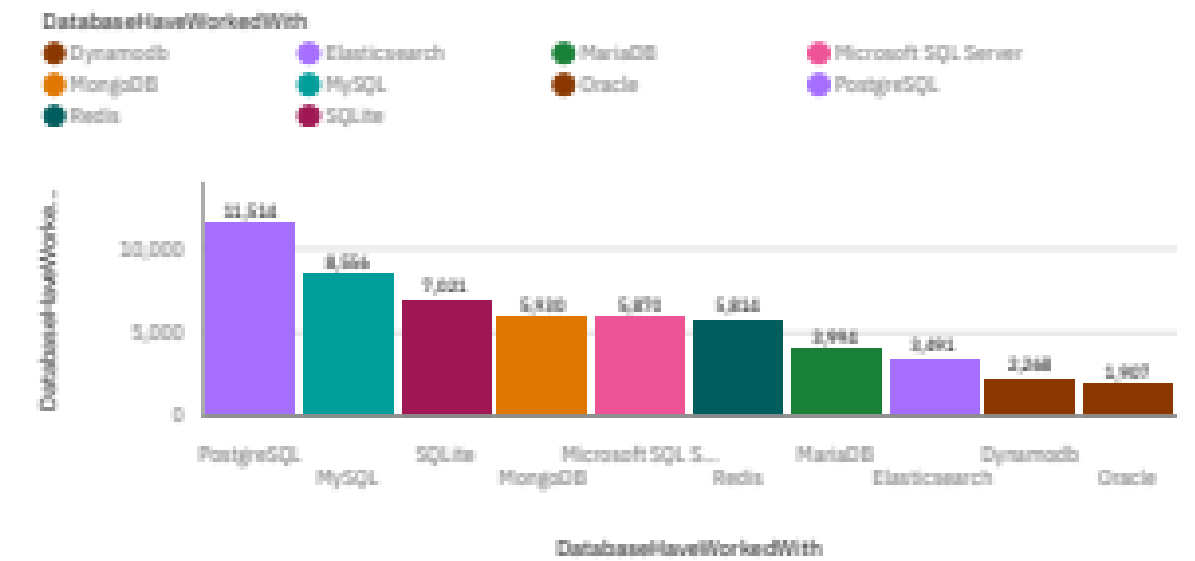
Current Technology Usage

Current Technology Usage

Top 10 Language Have Worked With



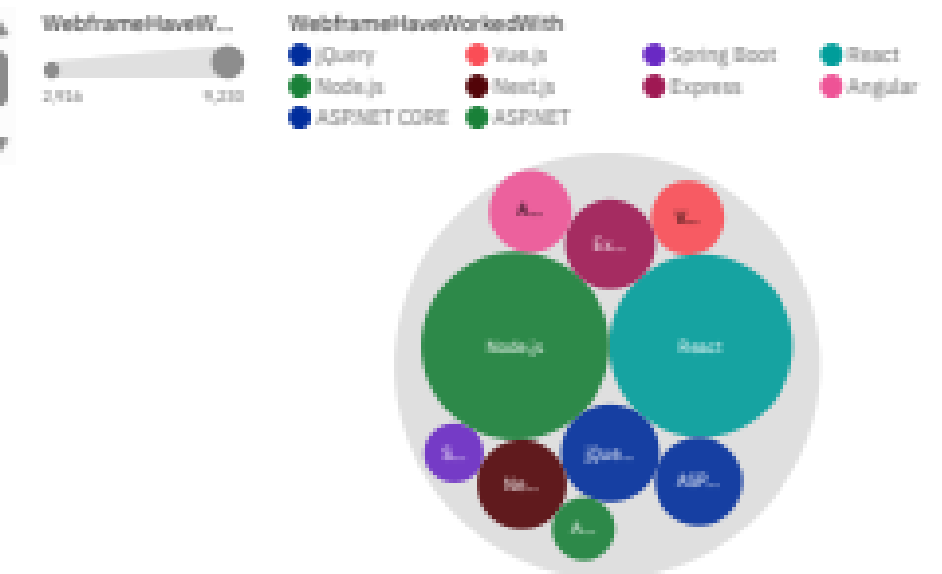
Top 10 Database Have Worked With



Platforms Mostly Used by Developers



Top 10 Web - Frames worked with



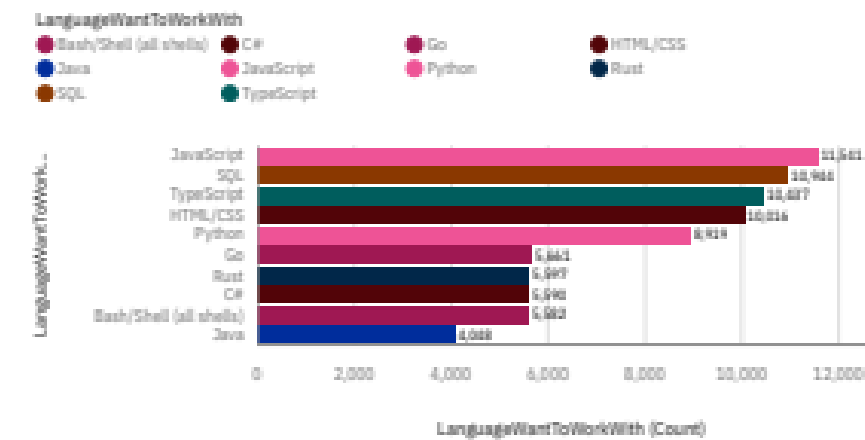


DASHBOARD

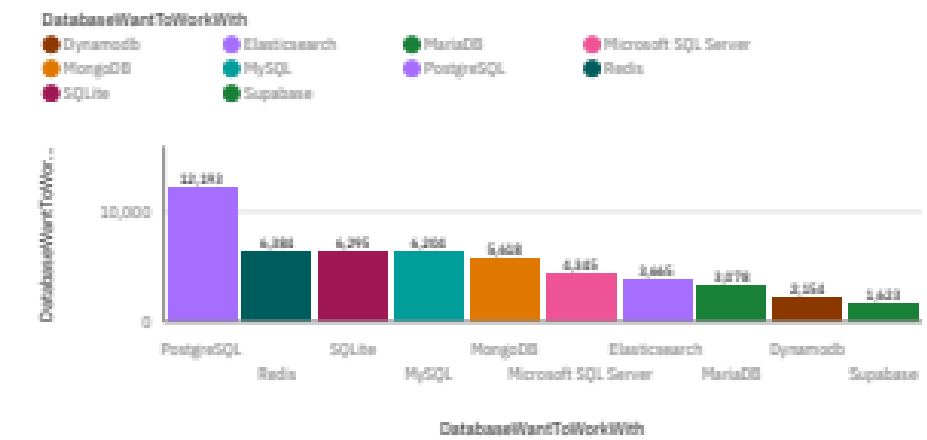
Future Technology Trend

Future Technology Trend

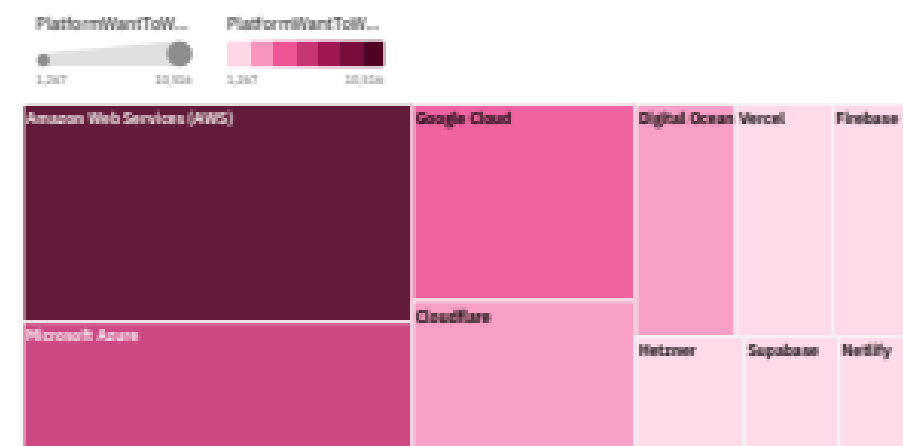
Top 10 Language Want To Work With



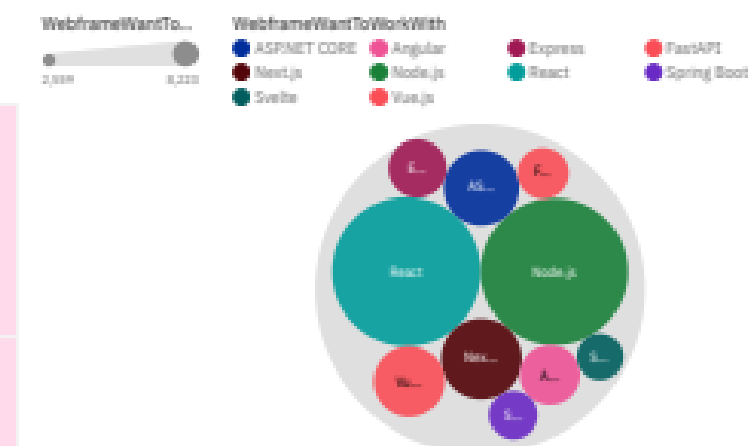
Top 10 Database Want To Work With



Platform Want To Work With



Top 10 Web-frame Want To Work With



Demographics

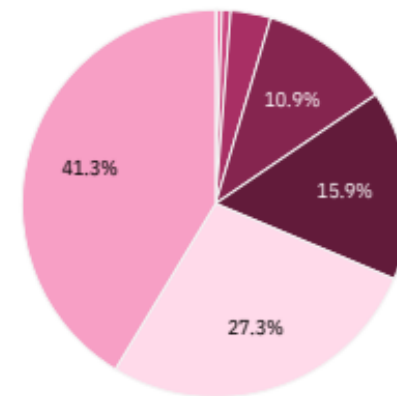
DASHBOARD

Demographics

Respondent Distribution by Age

Age

- Prefer not to say
- 18-24 years old
- 25-34 years old
- 35-44 years old
- 45-54 years old
- 55-64 years old
- 65 years or older
- Under 18 years old



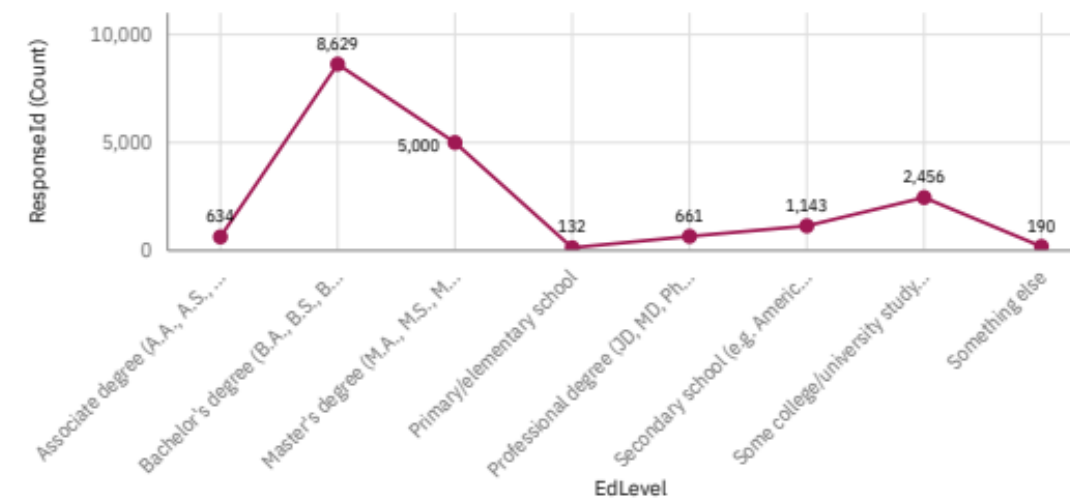
Respondent Count by Country

ResponseId (Count)

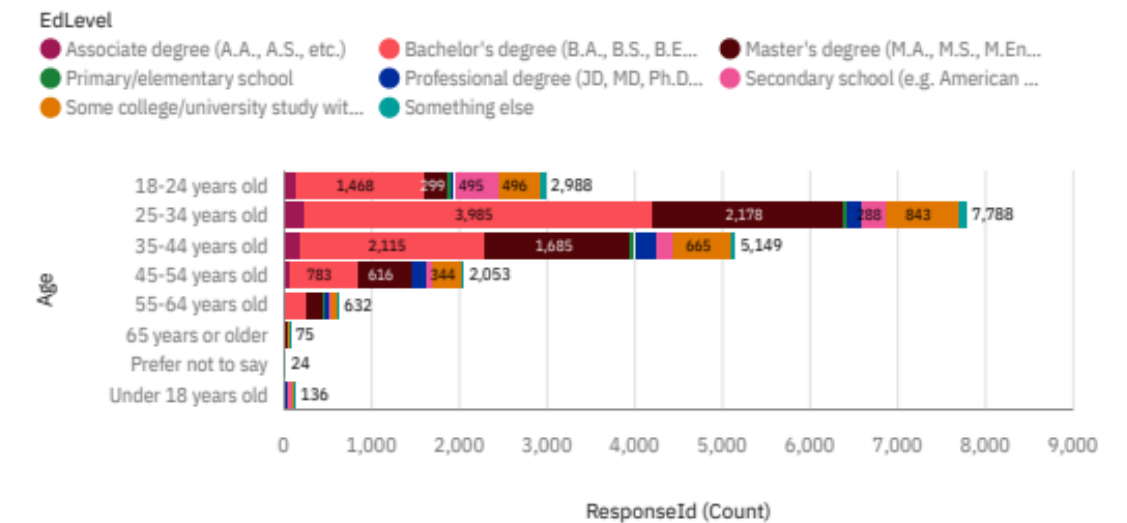
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Respondent Distribution by Formal Education Level



Education Level Distribution Across Age Groups



DISCUSSION

- **Programming & Databases:** JavaScript, SQL, and PostgreSQL remain the most in-demand, while TypeScript, Go, Rust, and Redis show rising popularity for the future.
- **Platforms:** AWS, Microsoft Azure, and Google Cloud are the most used, with Vercel and Supabase emerging as modern options.
- **Web Frameworks:** React, Node.js, and Next.js lead current usage; FastAPI and Svelte are gaining importance, while traditional frameworks like jQuery are declining.
- **Demographics:** Most respondents are from the USA, aged 25–34, and hold a Bachelor's degree.
- **Trends:** Demand is shifting toward modern technologies, new languages and frameworks, and more efficient cloud solutions.



OVERALL FINDINGS & IMPLICATIONS

Overall Findings:

- Core Skills Remain Essential: JavaScript, SQL, PostgreSQL, React, Node.js, and AWS dominate current usage.
- Rising Technologies: TypeScript, Go, Rust, Redis, FastAPI, and Svelte are gaining popularity.
- Cloud & Platforms: AWS, Azure, and Google Cloud continue to lead in usage.
- Web Development Shift: Modern frameworks like React, Node.js, Next.js are replacing traditional tools such as jQuery.
- Demographics: Majority of respondents are 25–34 years old, hold a Bachelor's degree, and are primarily based in the USA.

Implications for Industry:

- Developers need to maintain strong foundational skills while learning emerging tools.
- Organizations should invest in continuous upskilling to stay competitive.
- Focus on cloud and web-based solutions is critical for scalability and efficiency.
- Adoption of modern frameworks and APIs ensures faster development cycles and alignment with industry standards.
- Talent pipelines and training programs should consider age, education, and regional distribution to target upskilling effectively.



Conclusion

- Core skills (JavaScript, SQL, PostgreSQL, React, Node.js, AWS) remain in high demand.
- Emerging technologies (TypeScript, Go, Rust, Redis, FastAPI, Svelte) are gaining popularity.
- Continuous upskilling and adoption of modern platforms/frameworks are essential to stay competitive.